

1. The marriage rate per 1000 population for the years 1987-2009 is given by $M(x) = -0.137x + 11.068$, where x is the number of years after 1980.

- a. Why is this a linear function, with $y = M(x)$?
b. What is the slope? What does this slope say about the number of unmarried women who get married?

a. Choose the correct answer below.

- ☐ A. This a linear function because y is alone on the left side of the equation.
☐ B. This a linear function because x and y appear to the second power.
☐ C. This a linear function because x and y are not in separate terms.
☐ D. This a linear function because each of the variables x and y appear to the first power and it is written in the form $y = ax + b$.

b. What is the slope?

(Type an integer or a decimal.)

What does this slope say about the number of unmarried women who get married? Choose the correct answer below.

- ☐ A. It increased at a rate of 11.068 per thousand per year.
☐ B. It decreased at a rate of 0.137 per thousand per year.
☐ C. It decreased at a rate of 11.068 per thousand per year.
☐ D. It increased at a rate of 0.137 per thousand per year.

2. An electric board determines its monthly bills for residential customers by charging a base price of \$14.00 plus an energy charge of 13.21 cents for each kilowatt-hour (kWh) used. Write an equation for the monthly charge y (in dollars) as a function of x , the number of kWh used.

$y =$

(Type an expression using x as the variable. Use integers or decimals for any numbers in the expression. Do not include the \$ symbol in your answer.)

3. A business uses straight-line depreciation to determine the value y of a piece of machinery over a 10-year period. Suppose the original value (when $t = 0$) is equal to \$44,000 and its value is reduced by \$4400 each year. Write the linear equation that models the value y of this machinery at the end of year t .

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(Type an equation. Do not factor.)

4. A company finds it can produce 15 heaters for \$4550, while producing 20 heaters costs \$5900. Express the cost, y , as a linear function of the number of heaters, x .

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$y =$

(Type an expression using x as the variable. Simplify your answer. Do not factor.)

5. Suppose the monthly cost for the manufacture of golf balls is $C(x) = 2400 + 0.55x$ dollars, where x is the number of golf balls produced each month. Also, suppose the monthly revenue for the sale of golf balls is $R(x) = 1.15x$ dollars for x golfballs sold.

(A) The model for profit, including output and input units, would be:

$P(x) =$

☐ A. golfballs

for x

☐ A. dollars

☐ B. dollars

☐ B. golfballs

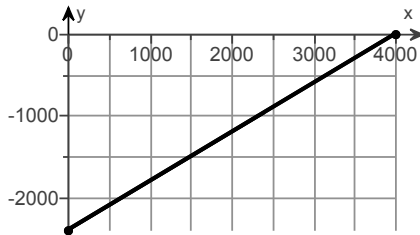
(B) The rate of change for profit is

☐ A. dollars per

☐ A. golfball

☐ B. golfballs

☐ B. dollar



y - intercept

x - intercept

(0, - 2400)

(4000,0)

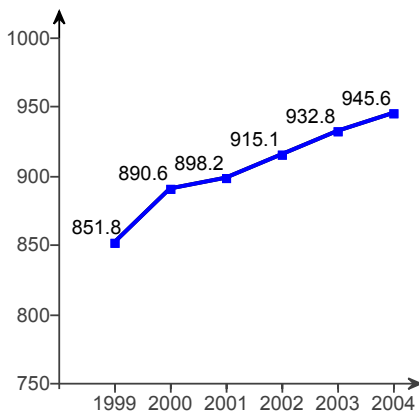
(C) The y - intercept tells us:

- ☐ A. When we sell 0 golfballs we lose 2400 dollars.
☐ B. When we sell 0 goldballs we get a 2400 dollar profit.
☐ C. We have to sell 2400 golfballs to break even.

(C) The x - intercept tells us:

- ☐ A. When we sell 0 goldballs we get a 4000 dollar profit.
☐ B. We have to sell 4000 golfballs to break even.
☐ C. When we sell 0 golfballs we lose 2400 dollars.

- *6. The graph below gives the number of millions of persons who took trips of 50 miles or more for the years 1999 through 2004.



Answer parts (a) through (c).

(a) Find the equation of the line which is the best fit for these data, with x equal to the number of years after 1999 and y equal to the millions of persons.

$y =$

(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

(b) Use the model to estimate the number of persons who took trips in 2008.

millions

(Round to two decimal places as needed.)

(c) In what year does the model estimate the number of persons who took trips as 986.48 million?

(Round to the nearest whole number as needed.)

7. The table shows the national expenditures for health care (in billions of dollars) for selected years from 2002 and projected to 2024. (These data include expenditures for medical research and medical facilities construction.)

Complete parts (a) through (c) below.

Year	Expenditures	Year	Expenditures
2002	1601	2014	3081
2004	1856	2016	3403
2006	2114	2018	3786
2008	2415	2020	4274
2010	2604	2022	4825
2012	2817	2024	5425

- a. Find a linear function that models the data, with x equal to the number of years after 2000 and y equal to the expenditures for health care in billions of dollars. Report the model with 3 significant digits.

$$y = \boxed{}x + \boxed{}$$

- b. What does the reported model predict the expenditures will be in 2050?

\$ billion (Use the answer from a to find this answer.)

- c. During what year does the reported model predict that the national expenditures for health care will be \$8069 billion?

The reported model predicts that the national expenditures for health care will be \$8069 billion in the year .

(Use the answer from a to find this answer.)